

Lab 11: Branch Prediction and Instruction Counting

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1 Introduction

This lab involves designing a branch prediction unit based on a Branch History Table (BHT) and integrating it within a simple 2-stage pipeline. The intention is to reduce delays caused by control hazards and to maintain an accurate record of instructions that successfully complete execution using an instruction counter.

2 Branch History Table (BHT)

The BHT is implemented using a **dual-port memory structure**, which enables concurrent operations:

1. **Read Operation (Port 1):** Retrieves stored branch behavior for the current instruction to assist in predicting the next program counter (PC).
2. **Write Operation (Port 2):** Updates the stored state based on the actual outcome of a previously executed branch in the second pipeline stage.

2.1 2-Bit Saturating Counter

Each BHT entry uses a 2-bit saturating counter to represent branch tendencies. Let $T = 1$ denote a taken branch and $T = 0$ denote not taken.

Table 1: 2-Bit Predictor State Transitions

Current State	Outcome (T)	Next State	Prediction
00 (Strongly NT)	0	00	0
00 (Strongly NT)	1	01	0
01 (Weakly NT)	0	00	0
01 (Weakly NT)	1	10	0
10 (Weakly T)	0	01	1
10 (Weakly T)	1	11	1
11 (Strongly T)	0	10	1
11 (Strongly T)	1	11	1

Let the state be represented as S_1S_0 . The prediction is derived from:

$$P = S_1 \tag{1}$$

Thus, the most significant bit determines whether the branch is predicted as taken or not.

3 Pipeline Operation and Branch Decisions

In the first stage of the pipeline, when a branch instruction is encountered, the prediction result determines the next PC value:

- If **Prediction = 1**: the next PC is set to $PC + br_imm$.
- If **Prediction = 0**: execution continues sequentially with $PC = PC + 4$.

3.1 Correction of Mispredictions

The predicted outcome is forwarded to Stage 2, where it is validated against the actual branch result. If a mismatch occurs, corrective steps are taken:

- If $Actual = 0$ and $Predicted = 1$: set $PC = prev_pc + 4$.
- If $Actual = 1$ and $Predicted = 0$: set $PC = prev_pc + s2_br_imm$.

In such cases, the incorrectly fetched instruction is discarded, and the pipeline experiences a stall while recovering the correct execution path.

4 Instruction Counter

The instruction counter tracks only those instructions that are successfully completed. Its behavior depends on the control signal *ready_stall*:

- If $ready_stall = 1$: the instruction is valid and committed, so the counter increments.
- If $ready_stall = 0$: the instruction is invalid (due to a stall or flush) and is not counted.